

# Information Reconciliation in Higher-Dimensional Quantum Cryptography

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**Abstract:** We report on a high-speed quantum cryptography system that utilizes simultaneous entanglement in polarization and time-bins. We also show a practical implementation of information reconciliation using concatenated low density parity check codes.

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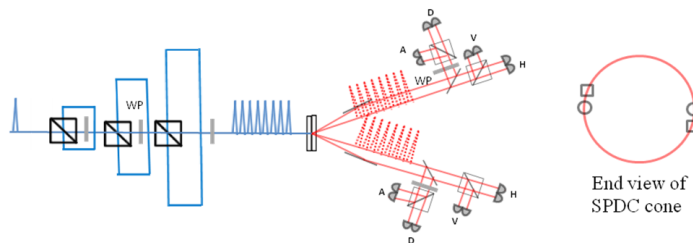
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## 1. Introduction

Entanglement of single photons allows for provably secure quantum communication useful for cryptography. There have been many quantum key distribution (QKD) experiments, and now even a few commercial systems. Unfortunately, the secret key rates for these systems are not usually large enough to encode things like video; contemporary QKD systems use single qubits, generating at most 1 bit of secret key per detection [2]. Instead of entangling photons in polarization alone (as in BB84), it is possible to entangle in additional degrees of freedom [1]. In particular, the timing of the photons can be exploited to encode additional bits per photon (bpp). For example, if each photon has the time range in which it can appear subdivided into 1024 segments (time bins), then a single detection event could in principle yield  $\log_2 1024 = 10$  bits of information [3]. Adding shared entropy from polarization can increase the information contained in a single detection event to 11 bits. The timing degree of freedom presents an interesting problem in ensuring that both parties (Alice and Bob) end up with the same secret key. Errors in photon detections require methods to reconcile differences between Alice and Bob's secret keys without giving away the keys themselves to a possible eavesdropper (Eve). The most commonly used information reconciliation protocol, cascade, cannot easily be adapted to timing degrees of freedom, partly due to the sparsity of detection events. In order to retain a maximal entropy, high performance error correction codes commonly used for radio communications can be adapted to the problem. In particular, low density parity check (LDPC) Slepian-Wolf codes are known to perform near the Shannon limit [4], making them ideal candidates for reconciling Alice and Bob's secret keys.

## 2. Optical Setup and Decoding

The source is described in figure 1. Inputs to the information reconciliation algorithm are a series of timestamps with accompanying channel information, corresponding to the 4 possible bases. Timestamps are temporarily stripped of polarization information, and are binned, allowing up to one detection per laser pulse. This sparse array is divided into segments of 1024 elements. Segments with no detected photons are assigned zeros, segments with one detected photon are assigned ones, and segments with multiple detections are assigned 2's. A non-binary LDPC code is run on segment numbers for Alice, and syndromes are sent to Bob, which enables Bob to reconstruct Alice's segment numbers without her explicitly revealing them. Bob then sends Alice information on which of their segments coincided. The most important case is when both received one photon detection in a 1024-segment. These 1-1 frames contain the majority of the system's shared entropy. Each 1-1 frame is assigned a number corresponding to the time bin within the segment in which the photon was found. These numbers are concatenated to create an array. Another non-binary LDPC code is run, allowing Bob to reconstruct Alice's photon-bin numbers. These bin numbers are then combined with entropy from other (1-2,2-2,etc) segmentations, as well as polarization encoding. After privacy amplification to account for entropy sent, these bits end up as a shared secret key.



### 3. Results

Current data is shown in the table below. Singles and coincidence rates given are photon detection rates for all 4 HVDA channels for both Alice and Bob. Polarization bit error rate is the percentage of data where Alice and Bob measure different polarizations despite using the same basis. This represents the fraction of information that Eve potentially knows, and determines the amount of privacy amplification necessary for a secure key. Shared entropy per second is calculated directly from singles and coincidences. Preliminary implementation of codes includes only time-bin correction of single-photon events. This means that all entropy associated with segmentation of detection events is discarded. Despite this, current implementation of codes retains up to 60% of the theoretical maximum entropy (dependent on data rate) associated with photons when run on timestamps from the time-to-digital converter.

|                                    | High Data Rate (High Power) | High Bits Per Photon (Low Power) |
|------------------------------------|-----------------------------|----------------------------------|
| Singles Rate                       | 5.8 MHz                     | 150 kHz                          |
| Coincidence Rate                   | 1.9 MHz                     | 45 kHz                           |
| Bit Error Rate (Polarization)      | 0.8%                        | 0.4%                             |
| Shared Entropy Per Second          | 10.4 Mbits                  | 470 kbits                        |
| Bits Per Coincidence               | 5.5 bpp                     | 10.4 bpp                         |
| Shared Entropy Given Current Codes | 6 Mbits/s                   | 290 kbits/s                      |

## 4. Conclusion

We have demonstrated a high-speed quantum cryptography system that uses photons with multiple degrees of freedom to operate at a hypothetical secure key rate of 10.4 Mbits/s. We can generate over 10 bits of shared entropy per coincidence. We also implemented information reconciliation using LDPC codes. Current work focuses on improved detectors for the optical setup, and optimization as well as further implementation of error correction codes.

## References

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